

CÁLCULO DE PRIMITIVAS: cadro resumo de algúns dos cambios de variable máis comúns e fórmulas trigonométricas útiles.

Tipo de integral (R función racional)	Cambio de variable ou fórmula trigonométrica aplicable	Exemplo:
$\int R(\operatorname{sen}(x), \cos(x)) dx$	Cambio universal: $t = tx\left(\frac{x}{2}\right)$, $dx = \frac{2}{t^2+1} dt$ $\operatorname{sen}(x) = \frac{2t}{t^2+1}$, $\cos(x) = \frac{1-t^2}{t^2+1}$	$\int \frac{\cos^3(x)}{\operatorname{sen}^4(x)} dx$
$\int R(\operatorname{sen}(x), \cos(x)) dx$	$t = \operatorname{sen}(x)$ ou $t = \cos(x)$	$\int \frac{\cos(x)}{\operatorname{sen}^2(x)} dx$
$\int R(\operatorname{sen}(x)) \cos(x) dx$	$t = \operatorname{sen}(x)$	$\int \operatorname{sen}^2(x) \cos(x) dx$
$\int R(\cos(x)) \operatorname{sen}(x) dx$	$t = \cos(x)$	
$\int R(tx(x)) dx$	$t = tx(x)$, $dx = \frac{dt}{1+t^2}$	$\int tx^3(x) dx$
$\int \operatorname{sen}^m(x) \cos^n(x) dx$		
$\int \operatorname{sen}^m(x) \cos^{2p+1}(x) dx$	$t = \operatorname{sen}(x)$	
m, n pares e $m, n \geq 0$	$\begin{cases} \operatorname{sen}^2(x) = \frac{1-\cos(2x)}{2} \\ \cos^2(x) = \frac{1+\cos(2x)}{2} \end{cases}$	$\int \operatorname{sen}^2(x) \cos^2(x) dx$
m, n pares e $m < 0$ ou $n < 0$	$t = tx(x)$, $dx = \frac{dt}{1+t^2}$	$\int \frac{\operatorname{sen}^4(x)+1}{\cos^2(x)} dx$
$\int \cos(px) \cos(qx) dx$	$\cos(px) \cos(qx) = \frac{1}{2}(\cos(p+q)x + \cos(p-q)x)$	
$\int \operatorname{sen}(px) \operatorname{sen}(qx) dx$	$\operatorname{sen}(px) \operatorname{sen}(qx) = \frac{1}{2}(-\cos(p+q)x + \cos(p-q)x)$	$\int \operatorname{sen}(6x) \cos(2x) dx$
$\int \operatorname{sen}(px) \cos(qx) dx$	$\operatorname{sen}(px) \cos(qx) = \frac{1}{2}(\operatorname{sen}(p+q)x + \operatorname{sen}(p-q)x)$	
$\int R\left(x, \left(\frac{ax+b}{cx+d}\right)^{\frac{m}{n}}, \dots, \left(\frac{ax+b}{cx+d}\right)^{\frac{r}{s}}\right) dx$	$t^k = \frac{ax+b}{cx+d}$, $k = m.c.m.(n, \dots, s)$	$\int \frac{6\sqrt{x}+1}{6\sqrt{x^7}+4\sqrt{x^5}} dx$
$\int R(x, \sqrt{x^2+a^2}) dx$	$x = a tx(t)$, $dx = \frac{a}{\cos^2(t)} dt$	$\int \frac{dx}{x^2\sqrt{1+x^2}}$
$\int R(x, \sqrt{x^2-a^2}) dx$	$x = a \sec(t)$, $dx = -a tx(t) dt$	
$\int R(x, \sqrt{a^2-x^2}) dx$	$x = a \operatorname{sen}(t)$, $dx = a \cos(t) dt$	
$\int R(x, \sqrt{ax^2+bx+c}) dx$ Se $a > 0$ Se $c > 0$, Se $ax^2+bx+c = a(x-\alpha)(x-\beta)$	$\sqrt{ax^2+bx+c} = \sqrt{ax} + t$, $x = \frac{t^2-c}{b-2\sqrt{at}}$ $\sqrt{ax^2+bx+c} = xt + \sqrt{c}$, $x = \frac{2t\sqrt{c-b}}{a-t^2}$ $\sqrt{ax^2+bx+c} = t(x-\alpha)$, $x = \frac{a\beta-\alpha t^2}{a-t^2}$	$\int \frac{dx}{x\sqrt{-x^2+5x-6}}$